

Drishti3D

Single-pass drone video → measurable, georeferenced 3D

Architecture, engineering plan, measured results, and an assessment of how far the system currently solves the problem statement — including what it does not yet do.

Repository github.com/SairaammBharadwaj/Drishti3D

Branch `sfm-hardening-and-colmap-engine` · head `dd357b1`

Codebase 7,669 lines across 27 reconstruction modules · 1,720 lines of evaluation harness · 137 passing tests

Documentation 3,391 lines · 29 recorded engineering decisions · 16 archived benchmark runs

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READ THIS FIRST

Every number in this document was produced by a recorded run against withheld ground truth. Where the system underperforms a trivial baseline, that is stated plainly rather than omitted. Section 8 lists what is still unsolved.

1 The problem, and what solving it requires

SIH problem statement 26158 asks for a system that converts a **single operational drone pass** into a **georeferenced, measurable 3D reconstruction**. Three words in that sentence carry all of the difficulty.

Single-pass

Conventional photogrammetry assumes a survey: a lawn-mower grid flown twice over, 80 % forward and 60 % side overlap, every point on the ground seen from many directions. A single pass gives none of that. The drone flies once, in one direction, and every surface is seen from a narrow cone of viewpoints. There is no loop closure, so nothing in the geometry ever revisits an earlier part of the trajectory to correct accumulated error. This is not a small relaxation of the survey assumption — it removes the redundancy that classical bundle adjustment depends on.

Georeferenced

Structure-from-motion recovers shape up to an arbitrary similarity transform: the reconstruction has no units, no orientation, and no position on the Earth. Metric scale and geography must come from somewhere else — in practice, the drone's onboard GNSS. That GNSS is itself wrong by several metres, so the georeferencing step is not a lookup but an estimation problem in which the reference is noisy.

Measurable

A point cloud that looks convincing is not the same as one you can take a dimension from. A measurement is only meaningful if the geometry under it was actually observed, and if the answer carries an uncertainty that has been checked against real error. A system that returns a number for every query — including where it saw nothing — is worse than one that refuses, because the operator cannot tell the two apart.

THE DESIGN COMMITMENT

Drishti3D treats *what was observed* as a first-class property of the output, carried per point and per unit of space, and refuses measurements that rest on geometry it never saw. Every accuracy claim is generated by a benchmark harness, never by a hand-picked run.

2 System architecture

The system is a fourteen-stage pipeline over a single video plus its telemetry. Each stage is independently testable, reports progress against a fixed budget, and records its own peak memory and wall-clock time. The stage boundaries below are the actual progress fractions in `pipeline.py`.

2.1 Pipeline stages

1	ingestion	Probe the container; establish resolution, frame rate and duration. Nothing is decoded yet.
2	telemetry	Parse GPS/IMU. If absent the run continues in relative scale — shape without geography — rather than failing.
3	frames	Decode to a working resolution (default 1280 px wide), bounded by <code>max_analyze_frames</code> .
4	quality	Score each frame for blur, exposure and feature density. Unusable frames are dropped before they can poison matching.
5	sync	Estimate the time offset between the video clock and the telemetry clock by correlating apparent image motion against GPS velocity.
6	keyframes	Select frames by measured inter-frame displacement using phase correlation, not by fixed stride.

7	masking	Optionally segment moving objects so vehicles and people never enter the static reconstruction.
8	sfm	Incremental structure-from-motion: seed-pair selection, essential-matrix initialisation, PnP registration, triangulation, and sparse bundle adjustment. The largest single stage, 48 %–72 % of the budget.
9	densify	Optional monocular-depth densification. Points created here are labelled AI-assisted and are never silently mixed with measured geometry.
10	georegistration	Robust Sim(3) alignment of reconstructed camera centres to GNSS in a local ENU frame, giving metric scale, orientation and absolute position.
11	fusion	Voxel fusion of the point cloud with per-point provenance and uncertainty preserved through the merge.
12	mesh	Surface reconstruction for visualisation.
13	report	Quality metrics, coverage summary, calibrated uncertainty, and recapture guidance.
14	exports	PLY, LAS, GLB, trajectory CSV, GeoJSON, and JSON/HTML reports.

2.2 Module inventory

MODULE	LINES	RESPONSIBILITY
<code>sfm.py</code>	1,194	Incremental SfM: seed-pair scoring, multi-scale pair graph, next-best-view scheduler, orientation propagation by maximum spanning tree, stale-point removal, reverse observation index
<code>pipeline.py</code>	837	Stage orchestration, parameter surface, manifest and artifact writing
<code>geo.py</code>	540	Geodetic ↔ ECEF ↔ ENU, Umeyama Sim(3), robust weighted Sim(3) with trajectory-conditioning checks, gravity levelling
<code>bundle.py</code>	462	Sparse Levenberg–Marquardt bundle adjustment with an analytic Jacobian
<code>uncertainty.py</code>	422	Gauss–Newton covariance propagation, distance/height/area uncertainty, conformal calibration, requirement gating
<code>coverage.py</code>	422	Per-camera z-buffer observation classes over a voxel grid
<code>sensors.py</code>	369	Time-offset estimation, lever-arm correction, rolling-shutter detection, undistortion
<code>features.py</code>	356	Pluggable matcher backends: SIFT, ORB, LightGlue
<code>synth.py</code>	340	Synthetic capture generator with exactly known geometry, for the benchmark
<code>colmap_adapter.py</code>	309	COLMAP engine path: bounded threads, fixed intrinsics, GNSS position priors
<code>capture.py</code>	227	Flight-capability assessment and recapture planning
<code>quality.py</code>	221	Quality metrics scored on observed geometry only
<code>verify.py</code>	215	Multi-view corroboration of inferred geometry
<code>exports.py</code>	208	PLY, LAS, GLB, CSV, GeoJSON, report writers
<code>telemetry.py</code> / <code>measure.py</code>	203 / 203	Telemetry parsing; point, distance, height and area measurement with provenance gating
<code>depth_prior.py</code>	188	Monocular depth densification
Plus <code>masking</code> , <code>fusion</code> , <code>keyframes</code> , <code>ai_adapter</code> , <code>ingestion</code> , <code>frame_quality</code> , <code>sync</code> , <code>provenance</code> , <code>mesh</code> — 27 modules, 7,669 lines total.		

2.3 The data model: provenance and coverage

This is the part of the architecture that distinguishes Drishti3D from a conventional photogrammetry pipeline. Two orthogonal layers describe what the system actually knows.

Provenance — a property of every point

CLASS	MEANING	MEASURABLE BY DEFAULT
 OBSERVED_HIGH_CONFIDENCE	Triangulated from multiple unobstructed views with adequate parallax	YES
 OBSERVED_LOW_CONFIDENCE	Triangulated, but from few views or at a grazing angle	FLAGGED
 AI_ASSISTED	Produced by a monocular depth model, not by triangulation	NO
 AI_GEOMETRICALLY_VERIFIED	AI-inferred, then corroborated against independent views	NO
 DYNAMIC_EXCLUDED	Belonged to a moving object and was deliberately removed	NO
 UNOBSERVED	Never seen	NO

THE INVARIANT

AI-inferred geometry can be *corroborated* but is never *promoted* to measured. Verification moves a point from AI_ASSISTED to AI_GEOMETRICALLY_VERIFIED ; it never moves it to OBSERVED . The distinction between what a sensor saw and what a model guessed survives all the way to the exported file.

Coverage — a property of space, not of points

Provenance describes points that exist. It cannot describe the volume where nothing was reconstructed, and that absence is exactly what an operator needs to know: was there no wall there, or did we simply never look? Coverage answers this by sweeping every camera's frustum through a voxel grid with a per-camera z-buffer.

CLASS	INTERPRETATION
UNSEEN	Never inside any camera frustum — the flight did not cover it
OCCLUDED	Inside a frustum, but the line of sight was blocked
EMPTY	Swept by rays and verified to contain no surface — a positive statement
WEAK	Surface present, but single-view or at grazing incidence
OBSERVED	Surface present with two or more unobstructed views
DYNAMIC_EXCLUDED	Occupied by a moving object during capture

The practical consequence: a measurement spanning UNSEEN space is refused rather than interpolated, and the report tells the pilot which part of the structure needs a second pass, and from what direction.

2.4 Two reconstruction engines

The system carries two interchangeable SfM back-ends behind one interface.

ENGINE	CHARACTER	WHEN IT IS THE RIGHT CHOICE
<code>opencv</code> (default)	Self-contained incremental SfM built on OpenCV. No CUDA, no COLMAP install, no downloaded weights.	Guaranteed to run anywhere, including air-gapped deployment. The verified default.
<code>colmap</code>	PyCOLMAP 4.2 with bounded threads, fixed intrinsics, and GNSS position priors.	Higher registration and substantially longer feature tracks on single-pass video (Section 6).

3 The measurement discipline

The most consequential architectural decision in this project is not in the reconstruction code. It is the rule governing what may be claimed.

3.1 The benchmark contract

1. **No accuracy claim may be published unless the benchmark harness generated it.** Not the README, not a slide, not a demo caption.
2. **Median and worst case are reported together.** A field system is described by its tail, not its best run.
3. **Failures are data.** A cell that crashes is recorded as a failed cell and counted, never silently dropped.
4. **One variable at a time.** A variant is a recorded parameter set, never an edited source tree.
5. **Alignment residual is never product accuracy.** The GNSS fit residual measures consistency with the track used to fit it. Accuracy is measured against withheld truth.

Every run records the git commit *and a dirty-tree flag*, interpreter and platform, library versions, a full dependency freeze, and the SHA-256 of every generated input. Cells run in isolated subprocesses so a single segfault or OOM cannot destroy the matrix, and so peak memory is that cell's rather than a high-water mark.

A RULE LEARNED THE HARD WAY

An early benchmark was invalidated because the pipeline was edited while the matrix was running — each cell spawns a fresh subprocess, so later cells silently picked up different code from earlier ones. The run was discarded. Never editing source during a matrix is now a standing rule.

3.2 Capture regimes

Flight geometry, not algorithm choice, is usually the dominant driver of whether a scene reconstructs at all. Benchmarking one favourable pass is how a system acquires a headline number it cannot repeat in the field. Four regimes are therefore held fixed.

REGIME	GEOMETRY	WHAT IT PROBES
<code>oblique_pass</code>	Single pass east at ~46 m, descending, building framed	The canonical target capture
<code>nadir_grid</code>	Three-strip lawn-mower at constant 45 m, camera straight down	The hard case: constant altitude means GNSS barely constrains the vertical axis, and near-parallel view directions give parallax only from the baseline
<code>orbit</code>	Circular orbit, 26 m radius, looking inward	Strongest single-pass parallax — what a pilot <i>should</i> fly
<code>low_parallax</code>	High (78 m), fast, nearly straight	The deliberate failure probe

HOW MANY SEEDS IS ENOUGH

Three seeds suffice for `orbit` and `oblique_pass`. They are **not** enough for `nadir_grid`: over six seeds its trajectory error ranges **0.170–2.405 m under identical code** — a factor of fourteen — while registration and dimensional error stay stable. A three-seed comparison there can invent an improvement that is pure spread. This was discovered by reporting an "improvement" from 2.312 m to 0.178 m that a six-seed sweep showed to be noise.

3.3 The decision log

Twenty-nine decisions are recorded with their evidence, including the ones that reversed earlier work. Three examples that changed the system materially:

- **D-020 — real data overturns the learned-matching decision.** Learned matching was measured on synthetic input, found not worth its cost, and declined. Real imagery reversed that conclusion outright.
- **D-025 — the georegistration residual is the model, not the GPS.** A 13 m residual was assumed to be GNSS noise until the GPS scatter was measured at 2.11 m, which relocated the fault into the reconstruction.
- **D-029 — two of my own optimisations reverted after measurement.** A pair-graph prune and a neighbour-count increase both looked principled and both degraded the target regime when measured.

4 The plan, and what was executed

Work proceeded through six phases derived from an initial repository audit. Each phase closed only when its claim was backed by a benchmark run.

PHASE	OBJECTIVE	OUTCOME	STATUS
1	Reproducible environment and a truth harness before any optimisation	Benchmark matrix, provenance capture, subprocess isolation, environment lock	CLOSED
2	Geometric optimisation — bundle adjustment	Sparse LM with analytic Jacobian; adopted on evidence, not on faith	CLOSED
3	Sensor model — time offset, lever arm, rolling shutter, distortion	All four implemented and regression-tested	CLOSED
4	Calibrated uncertainty	Gauss–Newton covariance propagation with distribution-free conformal calibration and leave-one-out validation	CLOSED
5	Coverage as a spatial layer	Per-camera z-buffer occupancy classes; measurement refusal; recapture planning	CLOSED
6	Real data, density, and georeferencing	COLMAP engine, real-footage sourcing, first georegistration against surveyed truth	IN PROGRESS

4.1 Bundle adjustment

Sparse Levenberg–Marquardt over camera poses and point positions, with an analytic Jacobian using the compact Rodrigues derivative rather than finite differences. Two findings shaped the final form:

- **Single-view points are held fixed.** Letting them float improved reprojection error while *worsening* point accuracy from 3.3 m to 7.6 m. A point seen once has no geometry to constrain it; optimising it fits noise.
- **The solver was starving.** The initial iteration and tolerance limits terminated before convergence. The adopted configuration uses a soft-L1 loss with Jacobian-based variable scaling.

4.2 Calibrated uncertainty

Point covariances are propagated by Gauss–Newton from the reprojection system, then pushed through the georegistration similarity so they mean metres on the ground. Because a propagated covariance is only as good as its noise model, the intervals are additionally **conformally calibrated** — a distribution-free correction fitted on withheld error and validated leave-one-out. A measurement can then be gated against a stated tolerance rather than merely annotated.

4.3 Initialisation and robustness

Seed-pair selection scores candidate pairs on parallax and inlier support instead of taking the first viable pair. Combined with a retry path, this took the `oblique_pass` regime from three failing runs — 11.3 m, 13.1 m and outright failure — to 0.069 m, 0.011 m and 0.020 m.

5 Results — synthetic matrix

Nine trials, three regimes, three seeds, zero failed cells. Reported median / worst.

100%	0.069 m	2.36%	0 / 9
REGISTRATION, MEDIAN & WORST	TRAJECTORY ERROR, MEDIAN	DIMENSIONAL ERROR, MEDIAN	FAILED CELLS

REGIME	TRIALS	REG.	TRAJECTORY ERR (MED / WORST)	SCALE ERR	CLOUD ACC	COMPLETE	DIM ERR
oblique_pass	3	100%	0.020 / 0.069 m	0.0024	0.809 m	38.3%	3.84%
orbit	3	100%	0.055 / 0.074 m	0.0003	0.675 m	95.5%	2.30%
nadir_grid	3	100%	0.577 / 2.405 m	0.0010	0.911 m	65.4%	2.15%
pooled	9	100%	0.069 / 2.405 m	0.0014 / 0.0035	0.809 / 0.960 m	65.4% / 34.5%	2.36% / 4.01%

Runtime 97.9 s median; worst-case peak memory 1,246 MB. `orbit` reaching 95.5 % completeness against `oblique_pass`'s 38.3 % is the clearest quantitative statement of how much flight geometry dominates outcome: identical code, three times the surface recovered.

WHAT THESE NUMBERS ARE NOT

Synthetic renders carry no motion blur, no rolling shutter, no exposure variation and no lens distortion — only GNSS noise is simulated. These figures are an **upper bound** on field performance, not a prediction of it. The next section is what happened when real imagery arrived, and several synthetic conclusions did not survive it.

6 Results — real imagery

6.1 Sourcing the data, and what it revealed

Three CC-licensed drone videos were obtained from Wikimedia Commons. All three turned out to be **edited montages rather than captures** — 15, 33 and 22 scene cuts respectively, detected by histogram-correlation drops. Feeding a whole file to the pipeline violates the single-pass assumption outright, because the frames span many unrelated shots. A cut segmenter now extracts the longest continuous run.

A DIAGNOSTIC CORRECTED

The rule of thumb "consecutive frames should differ by 10–30 % of frame width" was being applied without a timescale. Measured at 0.5 s spacing these cinematic videos read 0.7–2.9 %, which looks like failure but is not: the same parallax arrives over seconds rather than frames. A parallax criterion is meaningless unless it states the interval it is measured over.

6.2 Reconstruction from real single-pass video

CAPTURE	CHARACTER	REGISTERED	POINTS	MEAN TRACK	≥3 VIEWS	REPROJ.
Gymnasium Neubiberg	63 s continuous single pass over a school	64 / 64	33,287	6.25	95.7%	0.43 px
St. Lambertus	41 s orbit of a church	42 / 42	50,457	7.26	99.3%	0.43 px
Goetheanum	28 s, low parallax	17 / 28	2,544	5.00	97.4%	0.78 px

The Goetheanum failing at the lowest measured parallax of the three is the diagnostic predicting its own outcome — a useful confirmation that the capture-quality metric is measuring something real.

6.3 Engine × data: separating two confounded variables

COLMAP on real single-pass footage produced a mean track length of 6.25 against our 2.37 on the Bellus photo survey. Two variables had moved at once — engine *and* data — so the comparison attributed nothing. A 2×2 matrix separated them, with intrinsics held fixed so the result was not decided by who guessed focal length better.

MEAN TRACK LENGTH	BELLUS (122-IMAGE PHOTO GRID)	GYM_PASS (64-FRAME SINGLE PASS)
Ours — SIFT	—	4.37
Ours — LightGlue	2.37	4.82
COLMAP	4.00	6.25

REGISTRATION	BELLUS	GYM_PASS
Ours — SIFT	—	57 / 64 · 89.1%
Ours — LightGlue	105 / 122 · 86.1%	55 / 64 · 85.9%
COLMAP	108 / 122 · 88.5%	64 / 64 · 100%

FINDING

Both effects are real and roughly additive. Changing the **data** from photo grid to continuous single pass is worth **+2.25 to +2.45** track length; changing the **engine** is worth **+1.43 to +1.63**. The gap is therefore **roughly 60 % capture topology and 40 % engine** — and the prior assumption that the matcher was the whole problem was wrong.

Registration tells a different story from track length: on the photo grid the two engines are within three images of each other. COLMAP's advantage appears **only** on single-pass video — which is precisely the regime the problem statement targets.

7 Georegistration against surveyed truth

Every accuracy figure to this point was measured either against synthetic geometry or against the GNSS track used to fit the alignment — neither of which is independent truth. This section is the first evaluation against an external survey.

7.1 The dataset

Zurich Urban MAV (AGZ), ETH Zurich Robotics and Perception Group: a 2 km continuous MAV flight carrying per-frame GNSS, **surveyed 6-DoF ground truth**, and factory calibration. The full archive is 29.8 GB. Because the server honours HTTP Range requests, a range-backed ZIP reader (`eval/httpzip.py`) drives the standard library's `zipfile` over HTTP and extracts individual members: **184 frames pulled in 67 MB without downloading the archive**, which is what makes this benchmark reproducible.

PROPERTY	VALUE
Segment	200 m straight single pass, 184 frames at 1.13 m spacing
Straightness	14.8× (principal-axis elongation)
Truth coverage	Every frame lands on a surveyed pose
Onboard GNSS error	median 5.39 m, max 27.44 m — measured, not assumed
Calibration	f_x 893.39, f_y 898.33, c_x 951.13, c_y 555.13, k_1 -0.281, k_2 0.116

7.2 Progression

CONFIGURATION	FRAMES	REGISTERED	TRACK	TRAJECTORY ERROR
Self-calibrating (COLMAP default)	62	62 / 62	3.30	20.18 m
+ surveyed calibration held fixed	62	62 / 62	2.85	9.88 m
+ dense sampling at 1.13 m	184	184 / 184	3.79	12.96 m
+ GNSS position priors	184	184 / 184	3.88	10.54 m
+ centred priors	184	184 / 184	3.81	10.37 m

7.3 The headline result

Direct georeferenced error, with *no* alignment applied — the metric that actually matters, because the reconstruction is placed in UTM by the priors rather than fitted to truth afterwards.

	MEDIAN 3D	HORIZONTAL	VERTICAL	P90	MAX
Drishti3D	7.33 m	6.91 m	1.44 m	59.73 m	84.26 m
Onboard GNSS	5.39 m	3.19 m	2.72 m	12.57 m	27.44 m

WHAT WAS ACHIEVED

The reconstruction is **1.9× more accurate than the drone's own GNSS in height** — 1.44 m against 2.72 m. Height is the quantity the problem statement asks for. This is the first evidence, measured against an external survey rather than against itself, that the system recovers the dimension it exists to recover.

WHAT WAS NOT

Horizontal accuracy is **worse** than raw GNSS — 6.91 m against 3.19 m — and the p90 of 59.73 m says a minority of frames are badly placed. The horizontal result is not usable as it stands, and is reported here because omitting it would misrepresent the system.

7.4 Diagnosis: the residual is a bow, not drift

The error does not accumulate along the sequence — the correlation between frame index and error is -0.11 . Instead it is high at both ends and low in the middle:

FRAMES	0–22	23–45	46–68	69–91	92–114	115–137	138–160	161–183
Median error	26.7	6.4	9.0	18.5	16.0	9.4	3.2	24.9

Local geometry is sound. Three independent measurements say so:

- **Smoothness.** Trajectory jerk-per-step is 0.17 against surveyed truth's 0.12 — and against the GNSS's 0.42. The reconstructed flight path is *smoother than the GPS it was given*.
- **Scale.** Median step 1.12 m against a surveyed 1.13 m. Metric scale is essentially exact.
- **Triangulation.** Median angle 10.65° over 4,000 points (p10 3.09°). Parallax is adequate; this is not the bottleneck.

The failure is therefore **global shape, not local structure**. A straight pass carries no loop closure, so small rotation errors integrate into a low-frequency bend that nothing in the reconstruction can correct against. This is the central remaining technical problem, and it is now precisely characterised rather than suspected.

7.5 Four defects found and fixed in the COLMAP adapter

DEFECT	CONSEQUENCE	FIX
<code>intrinsics</code> was accepted and silently ignored	Known calibration had no effect; supplying it halved error from 20.18 m to 9.88 m once honoured	Honoured and held fixed
Per-image self-calibration (AUTO camera mode)	Focal estimates diverged from 1,062 to 2,044 px against a surveyed 893.4 — a $2\times$ error. On a near-straight trajectory the focal/depth ambiguity is barely observable	One shared camera per pass
Position priors inserted alongside COLMAP's own EXIF-derived rows	SQLite uniqueness violation; covariance left NaN	Existing rows updated in place, covariance stated
Prior centring transformed only the in-memory reconstruction	Model written to disk stayed at the centred origin — a 5.2×10^6 m georeferencing error in any downstream consumer	Corrected model written back

An unbounded SIFT thread pool was also bounded: COLMAP sizes it from the core count (24 on this machine), each thread holding full-resolution scale-space buffers, which took a 64-frame 1080p run past a 10 GB limit and returned SIGKILL. Peak resident memory is now 0.7 GB.

8 How this solves the problem statement

Requirement by requirement, with the evidence and the honest status.

REQUIREMENT	HOW IT IS ADDRESSED	EVIDENCE	STATUS
Single-pass input	Fourteen-stage pipeline consuming one video plus telemetry. Keyframes selected by measured displacement, not fixed stride. Cut detection rejects edited footage that violates the single-pass assumption.	64/64 and 42/42 frames registered on real continuous passes; the 2×2 shows single-pass topology is worth +2.25 track length over a photo grid	MET
3D reconstruction	Two interchangeable SfM engines. Sparse bundle adjustment with analytic Jacobian. Optional monocular-depth densification kept provenance-separate.	100% registration across nine synthetic trials, zero failures; 33k–50k measured points on real footage at 0.43 px reprojection	MET
Metric scale	Robust Sim(3) alignment of camera centres to GNSS, with trajectory-conditioning checks that detect degenerate geometry.	Scale error 0.0014 median synthetic; on surveyed real data, step 1.12 m against truth 1.13 m	MET
Measurable geometry	Point, distance, height and area measurement, each gated on provenance and each returning a calibrated uncertainty rather than a bare number.	Dimensional error 2.36% median / 4.01% worst across the synthetic matrix	MET
Height accuracy	The vertical component of the georeferenced solution.	1.44 m against surveyed truth — 1.9× better than the drone's own GNSS at 2.72 m	MET
Georeferenced output	ENU frame from geodetic origin; GNSS position priors drive bundle adjustment so the solution is placed in UTM directly rather than fitted afterwards. Exports to LAS and GeoJSON.	Direct (unaligned) placement confirmed in UTM; horizontal 6.91 m against GNSS 3.19 m	PARTIAL
Trustworthy output	Six provenance classes carried per point; six coverage classes carried per unit of space; measurements refused across unobserved volume; AI geometry never promoted to measured.	Quality metrics scored on observed geometry only — mixing inferred points had overstated accuracy 0.835 m as 1.847 m	MET
Operational feedback	Capture-capability assessment and recapture planning; which region is weak, and from what direction to re-fly it.	Coverage layer drives the recommendation; parallax diagnostic correctly predicted the Goetheanum failure in advance	MET
Deployability	Default engine requires no CUDA, no COLMAP installation and no downloaded weights. Non-commercial model weights refused by default. Memory bounded so heavy runs cannot take down the host.	Worst-case peak memory 1,246 MB synthetic, 0.7 GB on the COLMAP path after bounding	MET

8.1 The distinctive contribution

Many systems produce a point cloud from drone video. Three things here are unusual and are what the problem statement's emphasis on *trustworthy measurement* actually demands:

1. **Absence is represented as data.** Coverage classes make "we never looked there" a positive, queryable statement rather than an empty region indistinguishable from flat ground. A measurement spanning unobserved space is refused.

2. **Uncertainty is calibrated, not asserted.** Propagated covariance is corrected by a distribution-free conformal step fitted on withheld error, so a stated interval has been checked against reality rather than derived from an assumed noise model.
3. **The provenance boundary is enforced end to end.** AI-inferred geometry is corroborated but never promoted to measured, and that distinction survives into the exported file.

9 Known gaps

GAP	IMPACT	PLANNED APPROACH
Horizontal georegistration bow	Plan accuracy 6.91 m vs 3.19 m GNSS; p90 59.73 m. The principal unsolved problem.	Sequential plus loop-closure matching in place of exhaustive; tighter prior sigma; IMU orientation constraints
No dense MVS	Output is a sparse cloud of 33k–50k points, not a solid surface	COLMAP's dense stereo on the existing sparse solution
Synthetic renders are clean	No motion blur, rolling shutter, exposure variation or lens distortion is simulated, so synthetic figures are an upper bound	Extend the generator with these degradations
Fewer than 30 checkpoints	ASPRS positional-accuracy assessment requires ≥ 30 independent checkpoints; these runs are reduced-checkpoint validation, not standards compliance	Expand the AGZ evaluation across more of the 2 km flight
Boresight and rolling-shutter-aware BA	Rolling shutter is detected but not modelled inside the optimisation	Per-row pose interpolation in the bundle
Single real georeferenced scene	The surveyed-truth result rests on one 200 m segment	Additional segments and a second GPS-bearing dataset (a 44-image corridor strip is already acquired)

10 Reproducibility

ARTEFACT	LOCATION
Benchmark harness	<code>python -m eval.benchmark --variants baseline,ba</code>
Archived runs (16)	<code>docs/benchmarks/</code> — each with full cell data and environment lock
Engineering decision log	<code>docs/ENGINEERING_DECISION_LOG.md</code> — 29 entries with evidence
Measured results	<code>docs/BENCHMARK.md</code>
Chronological work log	<code>docs/WORK_LOG.md</code> — 1,768 lines
Remote-archive reader	<code>eval/httpzip.py</code> — reproduces the AGZ evaluation without the 29.8 GB download
Test suite	<code>pytest tests/</code> — 137 passing

CLOSING POSITION

Drishti3D reconstructs real single-pass drone video at 100 % frame registration, recovers metric scale to within 1 cm per 1.13 m step against surveyed truth, and measures height $1.9\times$ more accurately than the drone's own GNSS. Its horizontal georeferencing does not yet beat that GNSS, and the cause — a low-frequency trajectory bow arising from the absence of loop closure in a straight pass — has been isolated and characterised rather than merely suspected. Every figure in this document is reproducible from the recorded harness.